

CS 4782 Coding Assignment 3 Written Responses

Due: 3/21/25 11:59 PM on Gradescope

Late submissions accepted until 3/23/25 11:59 PM

Note: For homework, you can work in teams of up to 2 people. Please include your teammates' NetIDs and names on the front page and form a group on Gradescope. (Please answer all questions within each paragraph.) Please show all the relevant steps in your solutions.

Question 1:

How do the performances of the two models you just trained compare? What do you think might contribute to the differences you noticed? Write 2-3 sentences.

Answer 1:

We can find that the feature classifier model have a better performance than the pixel-space classifier. Because the feature-based classifier operate on high-dimensional representations that have been structured to maximize agreement between different views of the same image, but the pixel-space model looks at raw, flattened input.

Question 2:

How do the visualizations of the contrastive learning features compare to the visualization of the image pixels? How tightly clustered are the points for the different classes in each of the two visualizations? How might your observations relate to the utility of these two sets of features for image classification? Write 3-4 sentences.

Answer 2:

The contrastive learning features exhibit significantly tighter clustering and clearer semantic separation compared to the original image pixels, which appear as a disorganized and overlapping cloud of points. The pixel-space visualization is dominated by low-level noise and color distribution, the contrastive features demonstrate that the model has successfully learned to group images based on their actual object categories. In that case, we can observe that contrastive features are far more useful for image classification because they bridge the semantic gap, allowing a linear classifier to more easily define decision boundaries between classes.